

Comparative Analysis of MLP and Vision Transformers for Video Memorability Classification

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Abstract—This study compares Multi-Layer Perceptron (MLP) and Vision Transformer (ViT) models for video memorability classification. Using the *Memento10k* dataset, we reformulated memorability prediction from regression to binary classification, distinguishing low and high memorability videos based on quartile segmentation. We systematically evaluated traditional hand-crafted features such as HOG, HSVHist, LBP, and RGBHist against deep learning representations including AlexNet, DenseNet121, EfficientNet, ResNet50, and VGG using the same MLP classification architecture, and compared these with end-to-end ViT models processing raw video frames. To better understand temporal information, we extracted features from the first, middle, and last frames of each video and also evaluated a multi-frame approach combining them. This helped assess the contribution of different temporal segments across all models. The results show that deep learning features consistently outperform hand-crafted descriptors. Among traditional features, LBP achieved the highest F1-score (0.68). VGG achieved the highest F1-score among CNN-based features (0.83), while the ViT model achieved the highest accuracy with 0.86 using middle video frame inputs. Frame position analysis revealed that middle frames provide the most informative content across all architectures. The superior performance of ViT demonstrates the effectiveness of global attention mechanisms for capturing memorability-relevant visual patterns, establishing a new benchmark for video memorability classification.

Keywords—Video Memorability, Video Classification, Vision Transformers, Multimedia Information Retrieval.

I. INTRODUCTION

Digital platforms generate unprecedented volumes of short-form video yet only a fraction of this content leaves a durable trace in human memory. Video memorability, the probability that a viewer will spontaneously recall a clip after a delay, has therefore become a pivotal target for multimedia retrieval, targeted advertising, and data-driven instructional design [1].

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Accurate prediction of memorability would enable search engines to rank highlights more effectively, advertisers to maximize brand recall, and educators to curate materials that optimize long-term retention.

Previous research has predominantly treated memorability as a regression problem, mapping low-level descriptors such as color histograms and motion statistics to continuous recall scores. While this formulation captures subtle gradations, many real-world applications require categorical judgements, such as flagging “low” versus “high” memorability videos for recommender systems or content moderation pipelines. However, systematic evaluations of discrete classification remain scarce, and existing studies often rely on either hand-crafted features or deep representations, leaving open questions about their relative utility under controlled conditions.

This study addresses these gaps through a comprehensive two-tier comparative analysis. First, we benchmark traditional hand-crafted descriptors, including Histogram of Oriented Gradients (HOG), HSVHist, Local Binary Patterns (LBP), and RGBHist, against convolutional deep features from widely used architectures, including AlexNet, DenseNet121, EfficientNet, ResNet50, and VGG, within a uniform shallow Multi Layer Perceptron (MLP) framework. This design isolates the contribution of the representation while holding the classifier constant. Second, we evaluate an end-to-end Vision Transformer (ViT) model to assess whether its global self-attention mechanism provides measurable gains over sequential architectures using pre-extracted features.

Our investigation employs the publicly available *Memento10k* subset released for MediaEval 2023 Predicting Media Memorability task [2], framing memorability prediction as a binary classification problem that distinguishes “low” from “high” memorable content. We focus on individual key frames rather than full spatiotemporal sequences, reflecting practical constraints in low-latency retrieval systems and

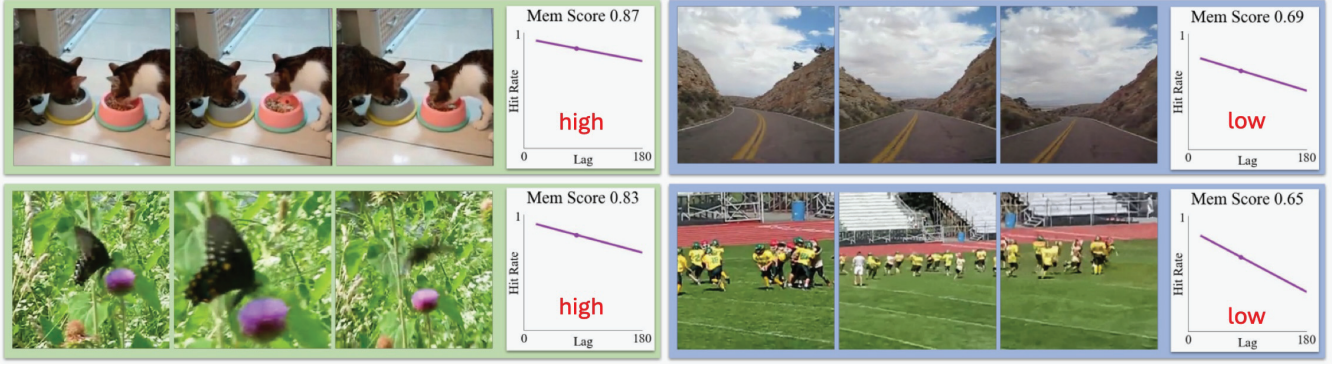


Fig. 1. Examples of *Memento10k* video clips annotated with human memorability scores and categories.

aligning with evidence that single frames often suffice for human recognition memory tasks.

The objectives of this research are to systematically address four key questions:

- How does the temporal position of video frames (first, middle, or last) affect classification performance across different feature extraction approaches?
- Does concatenating descriptors from multiple key frames improve accuracy over single-frame analysis?
- To what extent do deep features outperform traditional hand-crafted features across different frame positions?
- How does the end-to-end ViT compare with feature-based models in video memorability classification?

The rest of this paper is organized as follows: Section II describes the related work including datasets from previous studies and a description of the classifiers employed. Section III details the video reannotation process, feature representations, and classification architectures used in this study. Section IV presents the experimental results, and Section V summarizes the key findings.

II. RELATED WORK

In recent years, video memorability has emerged as a prominent and rapidly evolving research domain. Building on substantial progress in image memorability [3], research has increasingly shifted toward understanding the mechanisms underlying the memorability of dynamic visual content. Five major datasets have been developed for this purpose.

VideoMem is a large-scale benchmark comprising 10000 silent seven-second videos, developed by Cohendet et al. [1] to support systematic research on short-term and long-term video memorability. It provides robust memorability assessments.

Memento10k, introduced by Newman et al. [4], contains 10000 three-second videos annotated with short-term memorability scores and five textual descriptions (Fig. 1). Each video was evaluated by at least 90 participants, with variable repetition intervals (a few seconds to 10 minutes) enabling temporal analysis. Control repetitions served as quality checks without affecting final scores.

The *MediaEval 2020* dataset [5] includes 1500 six-second

videos from the Vimeo Creative Commons (V3C1) collection, providing both short-term and long-term memorability scores.

VIDEM [6] consists of 424 YouTube videos from 47 financial institutions, designed to analyze short- and long-term memorability and brand impact. Stratified random sampling ensured representative content types, and 1403 participants completed controlled recall tests.

LAMBDA [7] comprises 2205 YouTube ads from 276 brands across 113 industries. The average duration is 33 seconds. Data were collected in a controlled four-session study with 1749 participants, assessing memorability 24+ hours post-exposure via recognition and free recall tasks.

For this study, we selected *Memento10k* for its short, labeled videos, diverse content, and high annotation count, which facilitate supervised learning and enhance generalization across contexts.

Previous work [4] has primarily used regression models to predict average memorability scores, reporting correlation coefficients in the 0.60–0.70 range—insufficient for distinguishing low from high memorability cases. To address this limitation, we discretized regression scores into categorical classes via quartile segmentation, enabling a classification framework focused on differentiating low (*Q1*) and high (*Q4*) memorability videos.

This framework supports a range of feature extraction and modeling strategies. Classical methods such as HOG and HSVHist provide low-level descriptors (e.g., edges, color histograms) but have limited abstraction capability for complex tasks. In contrast, deep learning approaches using architectures such as AlexNet, DenseNet121, EfficientNet, ResNet50, and VGG produce semantically rich, discriminative features.

Recent studies show that combining modalities such as text, visual, and motion features can significantly improve episodic video memorability prediction [8], [9]. Beyond multimodal fusion, modular and multi-scale architectures have gained attention; for example, Dumont et al. [10] proposed a tiered model integrating low-, mid-, and high-level features to improve accuracy on *VideoMem* and *Memento10k*.

To leverage such features, classifiers like fully connected layers, logistic regression, and shallow MLPs are often used

TABLE .: OVERVIEW OF THE VISUAL FEATURES USED FOR VIDEO MEMORABILITY CLASSIFICATION

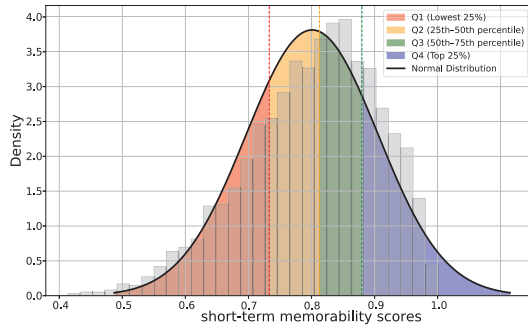
Feature	Description	Content
HOG [11]	Edge orientation	Shapes and memorable object contours
HSV Histogram [12]	Hue, saturation, brightness info	Vivid and bright colors
LBP [13]	Local texture information	Complex textures and facial features
RGB Histogram [12]	Basic color distribution	Colorful and emotionally striking scenes
AlexNetFC7 [14]	Deep feature vectors	Critical object and scene information
DenseNet121 [15]	Deep feature vectors	High-level feature representations
EfficientNet [16]	Optimized deep features	High performance with fewer parameters
ResNet50 [17]	Deep feature vectors	Deep-layered representations and detailed scene information
VGG [18]	Object and scene representations	Attractive scenes and objects

for their balance of efficiency and expressive power. Our study adopts similar approaches to ensure predictive accuracy and interpretability, while addressing challenges such as ambiguous visuals and mid-range scores, particularly in benchmarks like MediaEval 2023 [19].

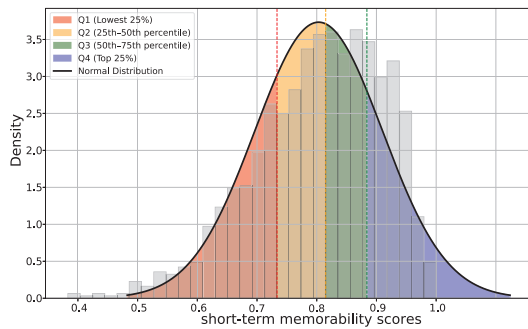
III. METHODOLOGY

A. Data Reannotation

In this study, the original memorability scores provided in the *Memento10k* dataset, represented as continuous values between 0 and 1, were systematically reannotated. These initial labels framed the task as a regression problem, with each video assigned a specific memorability score. However, the primary objective of our research was to investigate the distinction between videos with low and high memorability and to analyze the discriminative features underlying this difference. To enable a more interpretable and focused classification task, we discretized the continuous memorability scores into categorical classes based on quartile segmentation.



(a) Training set



(b) Test set

Fig. 2. Normal distribution and quartile segmentation of short-term memorability scores.

As shown in Fig. 2, all video scores were divided into four

quartiles ($Q1$, $Q2$, $Q3$, $Q4$):

- Videos in the first quartile ($Q1$) were labeled as “low” memorability,
- Videos in the second and third quartiles ($Q2$ and $Q3$) were labeled as “medium” memorability,
- Videos in the fourth quartile ($Q4$) were labeled as “high” memorability.

These newly defined categories were introduced as a new target value within the dataset. For the experiments, and in line with our central aim of distinguishing low and high memorability videos, only the $Q1$ (low) and $Q4$ (high) samples were included in the model training and evaluation; the medium ($Q2$ and $Q3$) instances were excluded from the study. This strategy has transformed the memorability regression problem into memorability binary classification task.

B. Data Preprocessing

The *Memento10k* dataset [4], used in this study, is publicly available for research purposes via official repositories. Access requires acceptance of the MediaEval 2023 Predicting Media Memorability task’s licensing terms [2]. The dataset comprises 7000 training and 1500 test samples; while memorability scores are provided for the training set, test labels are withheld and evaluated via official result submission. Given that the *Memento10k* dataset comprises YouTube videos collected over the past years, some of the original video links were no longer accessible due to removals from the platform. To ensure dataset integrity, these unavailable videos were systematically identified and excluded during the data cleaning stage of the preprocessing.

C. Feature Extraction

In this study, a diverse set of visual features was extracted to represent the visual content of video frames for memorability classification. These features were selected to capture both low-level perceptual attributes and high-level semantic representations, leveraging the complementary strengths of hand-crafted and deep features. Table I provides a detailed overview of the extracted features and their respective content characteristics. The top section lists hand-crafted features capturing low-level color, edge, and texture information. The bottom section presents deep features extracted from pre-trained CNNs, capturing high-level semantic content and scene representations.

Hand-crafted features were employed to describe low-

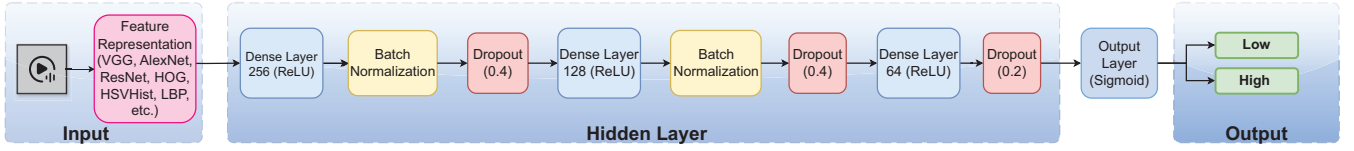


Fig. 3. Overview of the video memorability classification pipeline using the MLP model.

level visual properties known to influence human attention and memory. These include HOG [11], which encodes edge orientations and object shapes; LBP [13], which capture local texture variations; and color-based features such as RGB and HSV histograms [12], which reflect the distribution of colors and their perceptual qualities such as brightness and vividness.

To complement these, deep features were extracted from several widely adopted CNNs pretrained on ImageNet. These models include AlexNet [14], DenseNet121 [15], EfficientNet [16], ResNet50 [17], and VGG [18]. The activations from the fully connected layers of these networks were used as fixed-length feature vectors, capturing high-level abstractions. These representations are particularly useful for capturing concepts that contribute to a video’s semantic richness and its potential to be remembered.

D. Classification Architectures

MLP is a shallow feedforward neural network architecture composed of multiple fully connected layers, typically employing non-linear activation functions to model complex input-output relationships. Despite its structural simplicity compared to convolutional or transformer-based models, MLPs possess the capacity to learn intricate non-linear mappings, [20]. In this study, MLPs were employed as lightweight classifier heads applied to feature embeddings obtained from well-known deep feature extractors such as AlexNet, DenseNet, EfficientNet, ResNet50, and VGG, as well as from classical hand-crafted descriptors including RGBHist, HSVHist, LBP, and HOG. This architectural choice enables the prediction of video memorability scores in a computationally efficient manner. Owing to their compact structure and capacity to operate effectively within low-dimensional embedding spaces, MLP-based classifiers offer a favorable trade-off between predictive accuracy, computational cost, and interpretability. These attributes render MLPs well-suited for scenarios with constrained computational resources while maintaining adequate performance, thereby justifying their adoption in this work.

Fig. 3 demonstrates the architecture of the model used in this study. The model takes as input a feature vector extracted from precomputed representations. The MLP consists of three hidden dense layers with 256, 128, and 64 units, respectively, using ReLU activations, batch normalization, and dropout regularization. A final sigmoid output layer produces the probability of the input belonging to the “high” or “low” memorability class.

ViT represents a fully convolution-free architecture for image classification, achieved by adapting the Transformer

framework, originally developed for natural language processing, to the visual domain. In contrast to CNNs, which primarily capture local spatial patterns through convolutional filters, ViT leverages self-attention mechanisms to model global dependencies across image patches [21], [22]. This global receptive field enables ViT to capture holistic visual semantics, rendering it particularly effective in tasks where contextual integration is essential, such as video memorability prediction. An overview of the video memorability classification pipeline based on a fine-tuned ViT-Base-Patch16-224 model is illustrated in Fig. 4. When trained on sufficiently large-scale datasets, ViT has demonstrated competitive or superior performance relative to conventional CNN-based models, especially in domains that benefit from comprehensive visual abstraction. Moreover, ViT offers enhanced explainability through the visualization of attention distributions. These attributes collectively position ViT as a suitable and powerful architecture for this study, where the extraction of semantically meaningful representations from video frames is central to modeling memorability.

E. Evaluation and Explainable AI (XAI) Techniques

To assess the effectiveness of the proposed models, we conducted quantitative evaluations using standard classification metrics, including accuracy, precision, recall, and F1-score. Experiments were performed on a balanced dataset, with performance measured separately for classifiers utilizing hand-crafted and deep feature representations. Additionally, explainability metrics, Deletion AUC and Insertion AUC, were computed to evaluate the fidelity of visual explanations produced via Grad-CAM. This approach combines qualitative visual analysis with quantitative fidelity metrics to ensure the model’s predictions are both transparent and reliable.



Fig. 4. Overview of the video memorability classification pipeline using a fine-tuned ViT-Base-Patch16-224 model.

First, we utilized Gradient-weighted Class Activation Mapping (Grad-CAM) to generate visual explanations [23]. This technique produces saliency heatmaps that highlight the pixel regions most influential in the model’s classification decision for a given video frame. For the ViT architecture, we targeted the final transformer block (*encoder.layer[-1]*) to localize the features driving the prediction.

Second, to quantitatively validate the faithfulness of these heatmaps, we computed the Area Under the Curve (AUC) for two complementary metrics: Deletion and Insertion. The

Deletion metric measures the drop in the model's prediction probability as the most salient pixels (identified by Grad-CAM) are progressively removed from the image and replaced with a blurred baseline. A lower Deletion AUC indicates a more faithful explanation. Conversely, the Insertion metric evaluates the increase in prediction probability as the most salient pixels are incrementally introduced into the blurred baseline image. A higher Insertion AUC demonstrates that the identified regions are sufficient for the model to make its decision. This dual approach ensures a robust evaluation of the model's interpretability.

F. Experimental Setup

All experiments were conducted using Python 3.10 in the Google Colaboratory (Colab) environment, leveraging GPU acceleration (NVIDIA Tesla T4 and P100 via Colab Pro) for computational efficiency. Model implementation, training, and evaluation were entirely carried out within this environment. Neural network architectures, including ViT, MLP, and CNN-based models such as AlexNet, DenseNet, EfficientNet, ResNet, and VGG, were developed and trained using TensorFlow, PyTorch, and Keras.

IV. RESULTS AND DISCUSSION

The classification results across different feature types and temporal frames reveal a consistent pattern: deep learning features remarkably outperform hand-crafted features in video memorability classification tasks (see Table II). Among hand-crafted features, LBP and HSV histograms achieved relatively higher average F1-scores (0.66 and 0.66, respectively), whereas HOG and RGB histograms lagged behind. Deep features derived from pre-trained CNNs demonstrated

TABLE II. AVERAGE CLASSIFICATION PERFORMANCE OF DIFFERENT HAND-CRAFTED AND DEEP LEARNING FEATURES ACROSS ALL INPUT FRAMES (FIRST, MIDDLE, LAST) FOR VIDEO MEMORABILITY CLASSIFICATION

Feature	Accuracy	Precision	Recall	F1-Score
HOG	0.62	0.63	0.62	0.62
HSVHist	0.67	0.67	0.67	0.66
LBP	0.67	0.68	0.63	0.66
RGBHist	0.65	0.67	0.58	0.62
AlexNet	0.78	0.78	0.78	0.78
DenseNet121	0.75	0.75	0.75	0.75
EfficientNet	0.77	0.77	0.77	0.77
ResNet50	0.77	0.77	0.77	0.77
VGG	0.81	0.81	0.81	0.81

superior performance. VGG-based representations achieved the best results, with an average F1-score of 0.81, followed by AlexNet, ResNet50, and EfficientNet.

Frame-wise evaluation, reported in Table III), indicates that middle frames generally achieved higher classification accuracy across both hand-crafted and deep feature types. This trend aligns with prior findings suggesting that middle frames often capture salient content that represents the core semantics of a video clip. Moreover, AlexNet and VGG achieved their highest performance when using middle frames, with F1-scores of 0.80 and 0.82, respectively.

The use of concatenated frame-level features (Table IV)

TABLE III. PERFORMANCE COMPARISON OF HAND-CRAFTED AND DEEP LEARNING FEATURES ON FIRST, MIDDLE AND LAST VIDEO FRAMES FOR MEMORABILITY CLASSIFICATION

Feature	Frame	Accuracy	Precision	Recall	F1-Score
HOG	first	0.63	0.63	0.63	0.63
HOG	middle	0.62	0.63	0.62	0.62
HOG	last	0.62	0.62	0.62	0.62
HSVHist	first	0.66	0.66	0.66	0.66
HSVHist	middle	0.67	0.67	0.67	0.66
HSVHist	last	0.67	0.67	0.67	0.67
LBP	first	0.67	0.67	0.68	0.68
LBP	middle	0.65	0.65	0.63	0.64
LBP	last	0.68	0.72	0.59	0.65
RGBHist	first	0.64	0.68	0.53	0.59
RGBHist	middle	0.66	0.71	0.54	0.61
RGBHist	last	0.64	0.63	0.67	0.65
AlexNet	first	0.77	0.77	0.77	0.77
AlexNet	middle	0.80	0.80	0.80	0.80
AlexNet	last	0.77	0.77	0.77	0.77
DenseNet121	first	0.80	0.80	0.80	0.80
DenseNet121	middle	0.76	0.76	0.76	0.76
DenseNet121	last	0.71	0.71	0.71	0.70
EfficientNet	first	0.75	0.75	0.75	0.75
EfficientNet	middle	0.78	0.78	0.78	0.78
EfficientNet	last	0.77	0.77	0.77	0.77
ResNet50	first	0.77	0.77	0.77	0.77
ResNet50	middle	0.77	0.77	0.77	0.77
ResNet50	last	0.76	0.76	0.76	0.76
VGG	first	0.81	0.81	0.81	0.81
VGG	middle	0.82	0.82	0.82	0.82
VGG	last	0.80	0.80	0.80	0.80

provided a consistent increase in classification performance for both hand-crafted and deep features. For example, the F1-score for HOG features increased from 0.62 (average) to 0.72 when using concatenated frames. Similarly, VGG's F1-score improved from 0.81 to 0.83. This suggests that aggregating information across temporal positions captures complementary features critical for memorability prediction.

TABLE IV. PERFORMANCE COMPARISON OF DIFFERENT HAND-CRAFTED AND DEEP LEARNING FEATURES USING CONCATENATED REPRESENTATIONS FROM FIRST, MIDDLE AND LAST FRAMES FOR VIDEO MEMORABILITY CLASSIFICATION

Feature	Accuracy	Precision	Recall	F1-Score
HOG	0.72	0.72	0.72	0.72
HSVHist	0.69	0.69	0.69	0.69
LBP	0.69	0.70	0.69	0.69
RGBHist	0.67	0.68	0.67	0.66
AlexNet	0.81	0.81	0.81	0.81
DenseNet121	0.77	0.78	0.77	0.77
EfficientNet	0.78	0.78	0.78	0.78
ResNet50	0.8	0.8	0.8	0.8
VGG	0.83	0.83	0.83	0.83

The ViT model demonstrated the highest overall performance among all evaluated methods as shown in Table V. The model achieved its peak F1-score (0.86) when using middle frames, outperforming both hand-crafted and CNN-based models. This result highlights ViT's superior ability to extract global and semantically rich features through its attention-based mechanism, which appears particularly effective for modeling memorability.

To gain a deeper understanding of our ViT model's decision-making process, we analyzed its explainability using Deletion and Insertion AUC metrics. These metrics (see Fig. 5) provide insights into model behavior through Insertion and Deletion AUC scores. Both distributions are heavily right-skewed, with the majority of AUC values concen-

TABLE V: PERFORMANCE COMPARISON OF ViT MODEL USING DIFFERENT FRAME SELECTION STRATEGIES (FIRST, MIDDLE LAST AND FRAME-LEVEL CONCATENATION) FOR VIDEO MEMORABILITY CLASSIFICATION

Input Frame	Accuracy	Precision	Recall	F1-Score
first	0.85	0.85	0.85	0.85
middle	0.86	0.86	0.86	0.86
last	0.83	0.83	0.83	0.83
concatenation	0.84	0.84	0.84	0.84

trated below 0.2. The low deletion AUC values indicate that removing the most important regions, as identified by the model, substantially deteriorates prediction confidence, suggesting that the model is indeed relying on semantically meaningful content. Similarly, the Insertion AUC distribution confirms that adding informative regions improves model predictions incrementally. These distributions collectively affirm the alignment between learned representations and human-interpretable visual evidence, thereby supporting the model's reliability and providing an initial layer of interpretability.

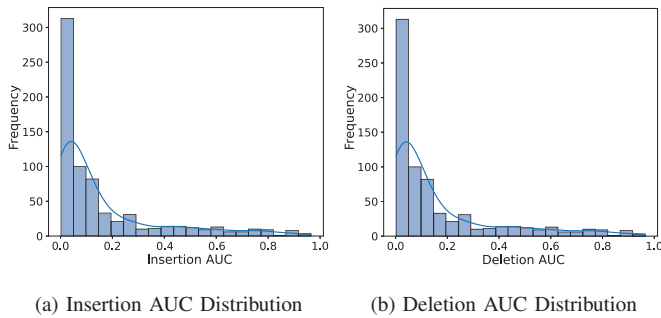


Fig. 5. Comparison of Deletion and Insertion AUC Distributions.

V. CONCLUSION

This study systematically evaluated the effectiveness of hand-crafted and deep visual features, alongside a fine-tuned ViT, in the task of video memorability classification. The results demonstrate that deep learning-based representations, particularly those derived from ViT and VGG, substantially outperform hand-crafted features across various temporal frames. Frame concatenation strategies further enhance performance, emphasizing the importance of integrating temporal context. Moreover, explainability analyses confirmed that the models rely on semantically meaningful regions when making predictions, enhancing their interpretability and trustworthiness. Overall, the findings highlight the potential of transformer-based architectures as robust and interpretable solutions for visual memorability prediction.

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